“Property finder”

Data Structures and Algorithms -3 (25CS2103E)



Abstract

The **Property Finder** project is a property search and discovery system designed to help users efficiently find suitable properties from a collection of property records. Real estate platforms may contain a large number of property listings, making manual searching difficult and time-consuming. Users commonly search properties using criteria such as location, price, property type, BHK, area, amenities, rating, and availability.

The core of the project is a **Java-based text corpus search system**. Property information is stored as text documents in a corpus, and the system searches these documents for user-entered patterns. The main DSA search implementation uses the **Knuth-Morris-Pratt (KMP) pattern matching algorithm**, which uses the Longest Proper Prefix which is also Suffix (LPS) array to avoid unnecessary character comparisons.

The project also contains a web-based Property Finder interface implemented using **HTML, CSS, JavaScript, and Node.js with Express**. The website provides a user-friendly interface for browsing properties, applying filters, viewing property details, shortlisting properties, and comparing selected properties.

The project repository also contains implementations and demonstrations of additional Data Structures and Algorithms concepts, including string algorithms, dynamic programming, graph and flow algorithms, approximation techniques, randomized algorithms, and parallel-processing-related algorithms. These components provide an extended DSA environment for academic demonstration and evaluation.

Overall, the Property Finder project demonstrates how algorithmic techniques can be applied to a practical real-estate search problem while combining a Java DSA implementation with a modern web interface.